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ASSIGNMENT

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ABSTRACT

Predictive maintenance is an emerging application of Artificial Intelligence (AI) and Machine Learning (ML) that helps industries monitor machine conditions and identify possible failures before they occur. In traditional maintenance approaches, machines are generally maintained at fixed intervals or repaired after a breakdown. These methods may lead to unexpected machine failures, increased maintenance expenses, production delays and equipment damage. Predictive maintenance provides a smarter approach by analysing machine operating conditions and predicting potential failures in advance.

The main aim of this project, **"AI-Based Predictive Maintenance System for Industrial Machines,"** is to develop a machine learning-based system that can identify whether an industrial machine is likely to experience a failure based on its sensor readings. The project uses the AI4I 2020 Predictive Maintenance Dataset, which contains 10,000 records and 14 attributes representing different machine operating conditions. Important parameters include air temperature, process temperature, rotational speed, torque and tool wear.

Initially, the dataset was examined through data preprocessing and Exploratory Data Analysis (EDA) to understand its structure, identify missing values, check duplicate records and study the distribution of machine failures. Five important sensor parameters were selected as input features for model development. Three machine learning algorithms, namely Logistic Regression, Decision Tree and Random Forest, were trained and evaluated using an 80:20 train-test split.

The experimental results showed that the Random Forest algorithm achieved the best overall accuracy of **98.15%**, with a failure-class precision of 73.13%, recall of 72.06% and F1-score of 72.59%. Based on the overall performance, Random Forest was selected as the final prediction model. These results demonstrate the potential of machine learning in identifying possible machine failures using operational sensor data.

To make the system accessible and easy to use, a web-based application was developed using Streamlit. The application allows users to enter machine sensor values and obtain predictions such as "Failure Detected" or "No Failure". It also displays the estimated failure score, maintains prediction history, visualises sensor readings and provides an option to download prediction records as a CSV file.

In conclusion, this project demonstrates how Artificial Intelligence and Machine Learning can be applied to industrial predictive maintenance. The developed system can serve as a basic decision-support tool for identifying potential machine failures and improving maintenance planning. However, further testing with real industrial data is necessary before using the model in actual industrial environments.

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1. INTRODUCTION

Industrial machines play an important role in manufacturing and production processes. The proper functioning of these machines is essential for maintaining productivity, product quality and operational efficiency. However, unexpected machine failures can interrupt production, increase maintenance costs and cause equipment damage.

Traditional maintenance methods mainly depend on scheduled inspections or repairing machines after a breakdown. These methods may not identify problems at an early stage. Predictive maintenance provides an alternative approach by analysing machine operating conditions and identifying possible failures before they occur.

Artificial Intelligence and Machine Learning techniques can analyse sensor readings and identify patterns related to machine failures. This project develops an AI-based predictive maintenance system using machine learning algorithms to predict potential industrial machine failures.

2. PROBLEM STATEMENT

Unexpected industrial machine failures can cause production delays, financial losses and increased maintenance expenses. Traditional maintenance approaches may fail to identify early warning signs of equipment problems.

Therefore, there is a need for an intelligent system that analyses machine sensor data and predicts possible failures in advance. This project addresses the problem by developing a machine learning-based system that classifies machine conditions as failure or no failure.

3. OBJECTIVES

The main objectives of this project are:

- To analyse industrial machine sensor data.
- To perform data preprocessing and Exploratory Data Analysis.
- To identify important parameters associated with machine failures.
- To implement Logistic Regression, Decision Tree and Random Forest algorithms.
- To compare the performance of different machine learning models.

- To select the best-performing model for prediction.
- To develop an interactive web application using Streamlit.
- To provide machine failure predictions based on user-entered sensor values.

4. DATASET DESCRIPTION

- This project uses the **AI4I 2020 Predictive Maintenance Dataset**, which is designed for machine predictive maintenance research. The dataset contains 10,000 records and 14 attributes representing machine operating conditions and failure information.
- The dataset includes the following important parameters:

Attribute	Description
Air Temperature	Temperature of surrounding air
Process Temperature	Temperature during machine operation
Rotational Speed	Machine rotation speed in RPM
Torque	Rotational force measured in Nm
Tool Wear	Accumulated tool usage time
Machine Failure	Indicates whether machine failure occurred

- The target variable is **Machine Failure**, where 0 represents No Failure and 1 represents Failure.

5. DATA PREPROCESSING

- Data preprocessing is an important step in preparing the dataset for machine learning. It helps improve data quality and ensures that the input data is suitable for model development.
- The following preprocessing activities were performed:
- **1. Dataset Loading:** The dataset was imported into Python using the Pandas library.
- **2. Data Inspection:** The dataset shape, column names, data types and statistical information were examined.

- **3. Missing Value Checking:** The dataset was checked for missing values. No missing values were identified.
- **4. Duplicate Checking:** Duplicate records were checked to ensure data consistency. No duplicate records were found.
- **5. Feature Selection:** Five important sensor parameters were selected: Air Temperature, Process Temperature, Rotational Speed, Torque and Tool Wear.
- **6. Target Selection:** Machine Failure was selected as the target variable.
- **7. Train-Test Splitting:** The dataset was divided into 80% training data and 20% testing data using stratified sampling.
- These preprocessing steps prepared the dataset for training and evaluating the machine learning models.

6. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was performed to understand the dataset, identify patterns and examine the relationship between machine operating conditions and failures.

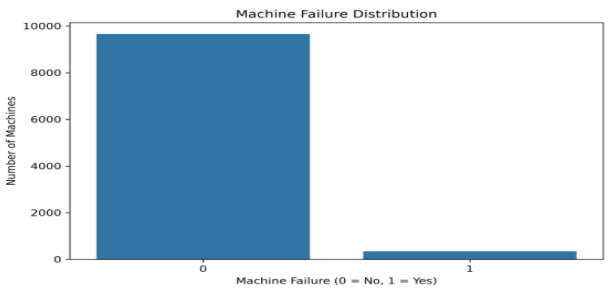
Python libraries such as Pandas, Matplotlib and Seaborn were used to analyse and visualise the dataset.

6.1 Machine Failure Distribution

The dataset contains 10,000 records. Among them, 9,661 records (96.61%) represent normal machine operation, while 339 records (3.39%) represent machine failures.

This shows that the dataset is imbalanced, with significantly more normal operating records than failure records.

Figure 2: Machine Failure Distribution



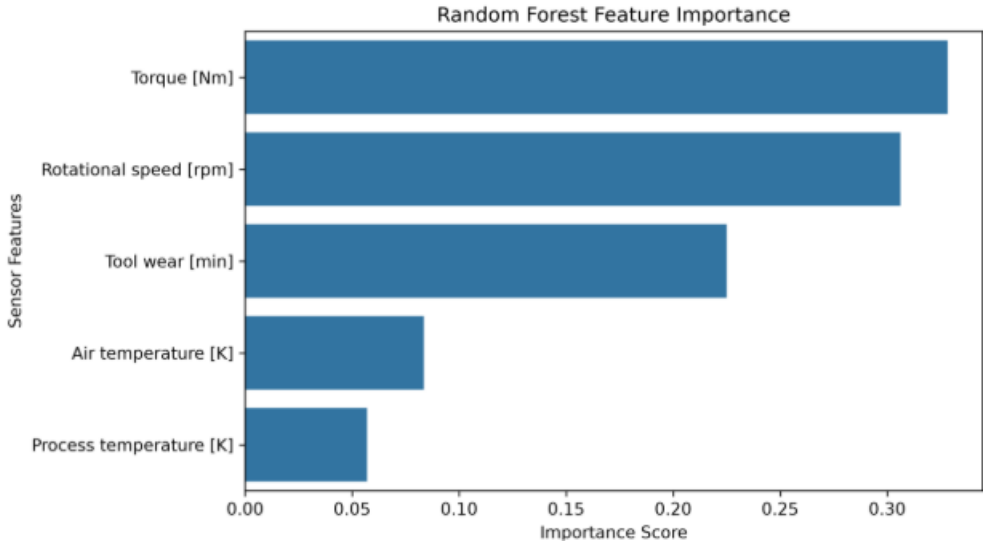
6.2 Sensor Parameter Analysis

The important sensor parameters analysed in this project include:

- Air Temperature
- Process Temperature
- Rotational Speed
- Torque
- Tool Wear

These parameters help understand machine operating conditions and provide useful information for predicting possible failures.

Figure 3: Sensor Parameter Distribution



6.3 EDA Findings

The analysis helped identify the imbalance in the target variable and understand the distribution of machine sensor readings. These findings supported feature selection and further machine learning model development.

7. MACHINE LEARNING ALGORITHMS

Machine Learning algorithms were used to develop a predictive model that identifies potential industrial machine failures based on sensor readings.

In this project, three classification algorithms were implemented and compared.

7.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts the probability of a machine failure based on the input sensor parameters.

7.2 Decision Tree

Decision Tree is a supervised learning algorithm that makes predictions using a tree-like structure of conditions. It divides the input data into smaller groups based on important features and produces a failure prediction.

7.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction performance. It generally provides better stability and reduces overfitting compared with a single decision tree.

7.4 Model Training

The dataset was divided into training and testing sets using an 80:20 ratio.

- Training Data: 8,000 records
- Testing Data: 2,000 records
- Input Features: 5 sensor parameters
- Target Variable: Machine Failure

All three algorithms were trained using the training dataset and evaluated using the testing dataset.

7.5 Model Selection The performance of all three algorithms was compared using accuracy, precision, recall and F1-score. Random Forest was selected as the final model because it achieved the best overall balance of evaluation metrics.

8. MODEL EVALUATION AND COMPARISON

The three machine learning algorithms were evaluated using the testing dataset. Different evaluation metrics were used to compare their performance.

8.1 Evaluation Metrics

- **Accuracy:** Measures the overall percentage of correct predictions.
- **Precision:** Measures how many predicted failures were actually failures.
- **Recall:** Measures how many actual failures were correctly identified.
- **F1-Score:** Represents the balance between precision and recall.

8.2 Model Performance Comparison

The following table presents the experimental results obtained from the models.

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82.10%	13.93%	82.35%	23.83%
Decision Tree	93.00%	31.00%	91.00%	46.00%
Random Forest	98.15%	73.13%	72.06%	72.59%

8.3 Result Analysis

Random Forest achieved the highest overall accuracy of 98.15% and the best F1-score of 72.59%.

Decision Tree achieved a higher recall of 91%, indicating that it identified a greater proportion of actual failures. However, Random Forest provided a better overall balance between precision and recall.

Therefore, Random Forest was selected as the final model for the Streamlit application.

Figure 4: Machine Learning Model Performance Comparison

```
model_comparison = pd.DataFrame({
    "Model": [
        "Logistic Regression",
        "Decision Tree",
        "Random Forest"
    ],
    "Accuracy (%)": [
        lr_accuracy * 100,
        dt_accuracy * 100,
        accuracy * 100
    ]
})

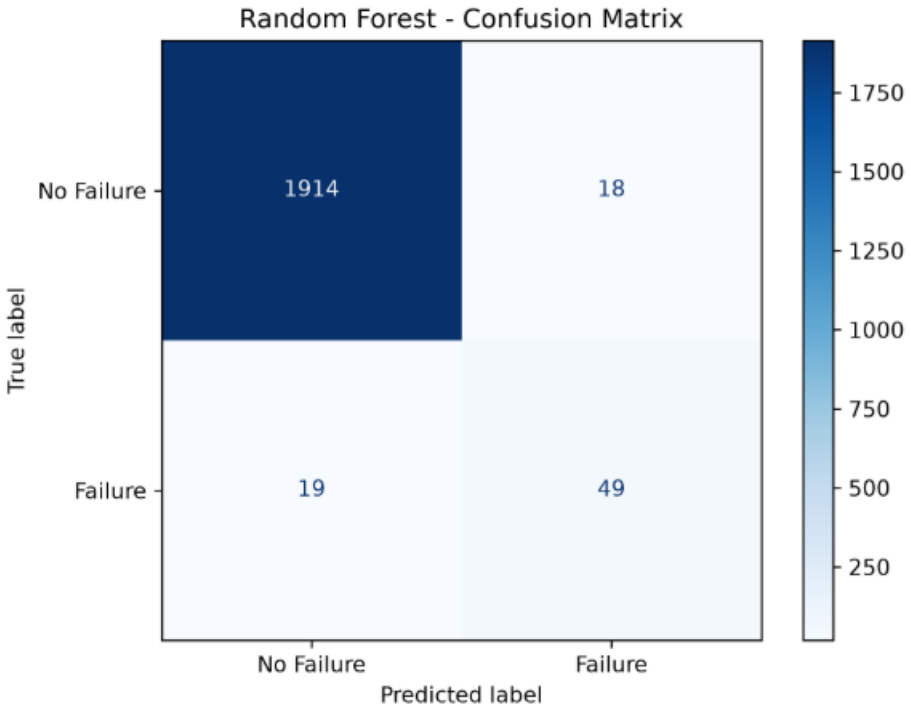
model_comparison = model_comparison.sort_values(
    by="Accuracy (%)",
    ascending=False
)

model_comparison
```

41]

	Model	Accuracy (%)
2	Random Forest	98.15
1	Decision Tree	92.75
0	Logistic Regression	82.10

Figure 5: Random Forest Confusion Matrix



9. STREAMLIT APPLICATION DEVELOPMENT

A web-based application was developed using Streamlit to make the predictive maintenance system interactive and user-friendly.

The trained Random Forest model was integrated into the application using the Joblib library. Users can enter machine sensor readings and receive predictions based on the selected input values.

9.1 Technologies Used

- **Python:** Main programming language.
- **Streamlit:** Used to develop the interactive web application.
- **Scikit-learn:** Used for machine learning model development.
- **Joblib:** Used to save and load the trained model.
- **Plotly:** Used for interactive data visualisation.
- **Pandas:** Used for data handling and processing.

9.2 Application Features

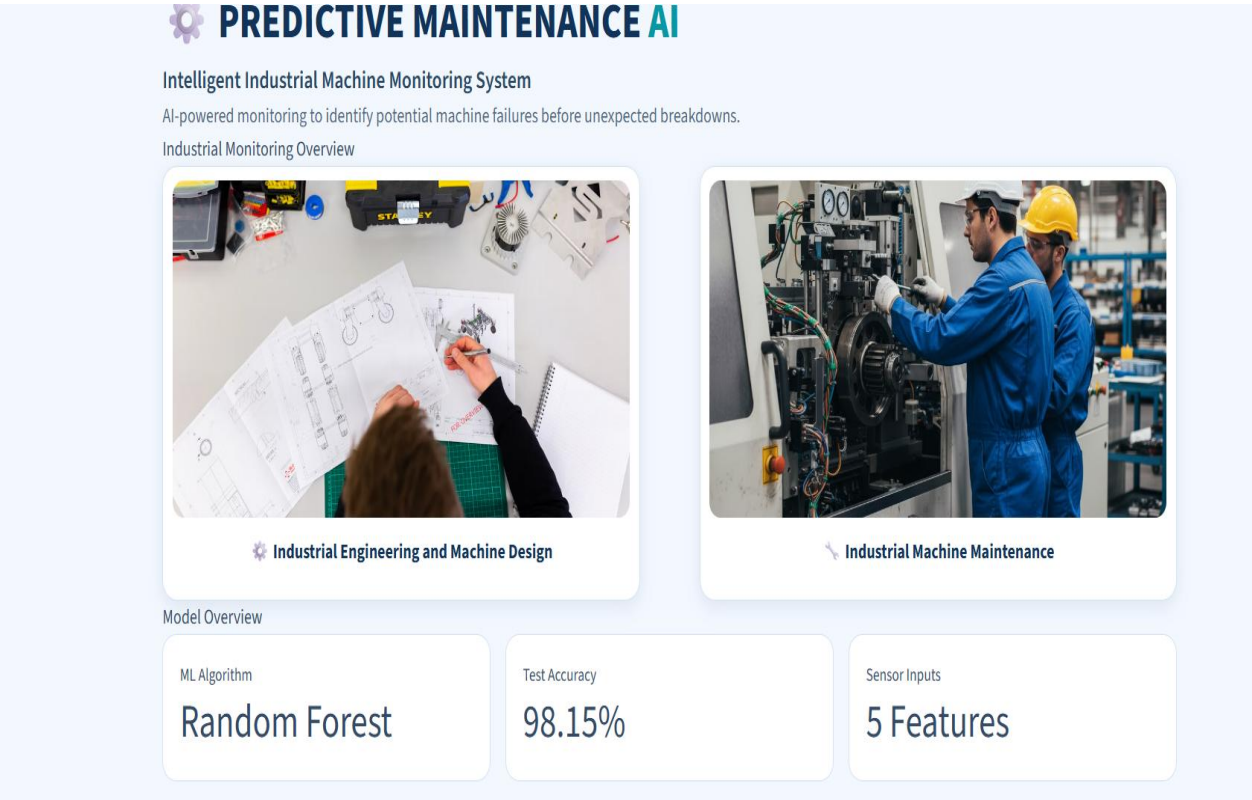
The application provides the following features:

1. User-friendly interface for entering machine sensor readings.
2. Prediction of machine failure or normal operation.
3. Display of model-estimated failure score.
4. Prediction history tracking.
5. Interactive sensor reading history graph.
6. CSV download option for prediction records.

9.3 Application Interface

The application was deployed using Streamlit Community Cloud, allowing users to access the system through a web browser.

Figure 6: Predictive Maintenance AI Application Home Page



10. MACHINE FAILURE PREDICTION RESULTS

The developed application accepts five machine sensor parameters as input and uses the trained Random Forest model to predict the machine condition.

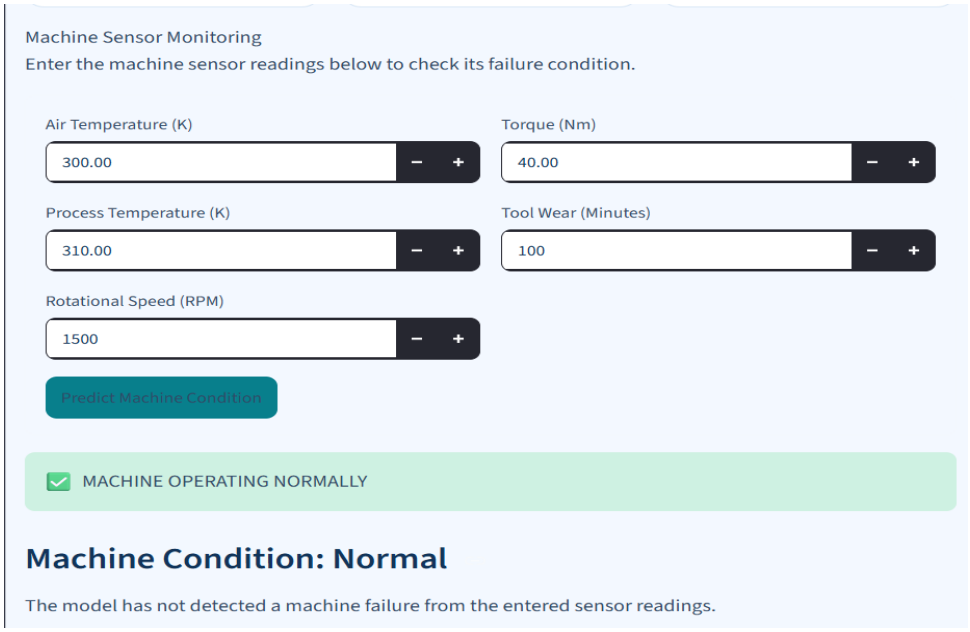
The system produces two possible prediction results.

10.1 No Failure Prediction

When the entered sensor values indicate normal machine operating conditions, the application displays **"No Failure"**.

This result indicates that the model has not identified a machine failure based on the supplied sensor readings.

Figure 7: No Failure Prediction

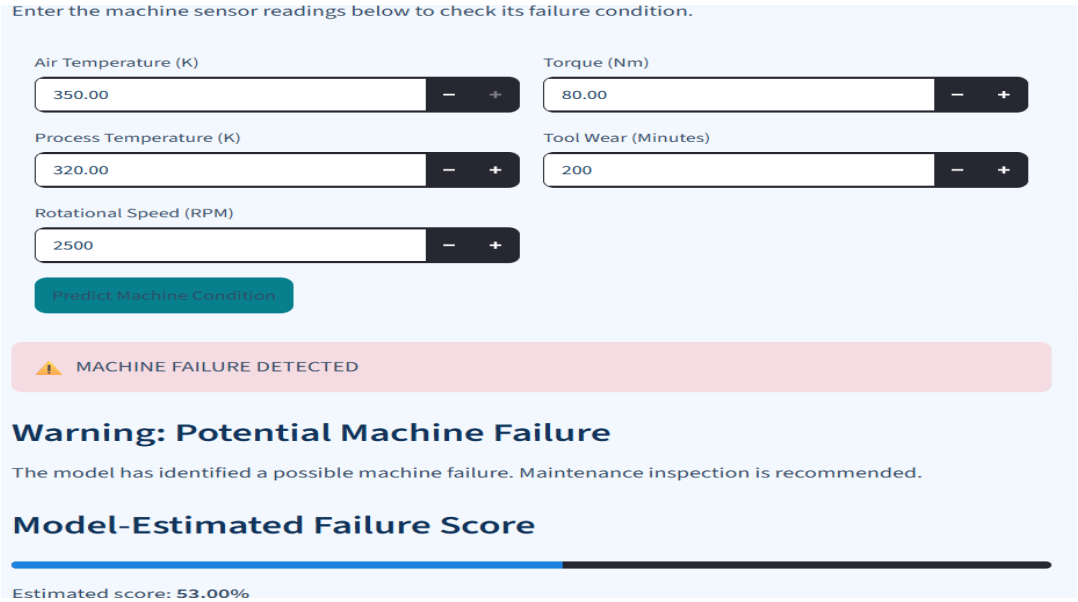


10.2 Failure Detected

When the entered sensor values indicate a potential failure pattern, the application displays **"Failure Detected"**.

This helps users identify machine conditions that may require further inspection or maintenance.

Figure 8: Failure Detected Prediction



10.3 Prediction Output

The application also displays a model-estimated failure score to provide additional information about the prediction.

These outputs demonstrate how the trained machine learning model can be used through a simple interactive interface.

11. PREDICTION HISTORY AND RESULTS

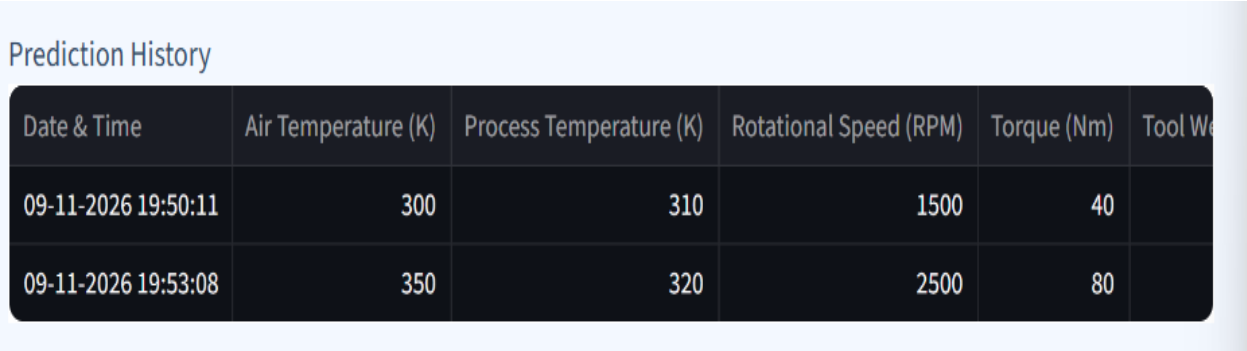
The developed application includes a prediction history feature that stores the sensor readings and corresponding prediction results during the application session.

11.1 Prediction History

The prediction history section allows users to review previously performed predictions. It helps users monitor the results of multiple machine sensor readings.

The application also provides an option to download the prediction history as a CSV file for future reference and analysis.

Figure 9: Prediction History

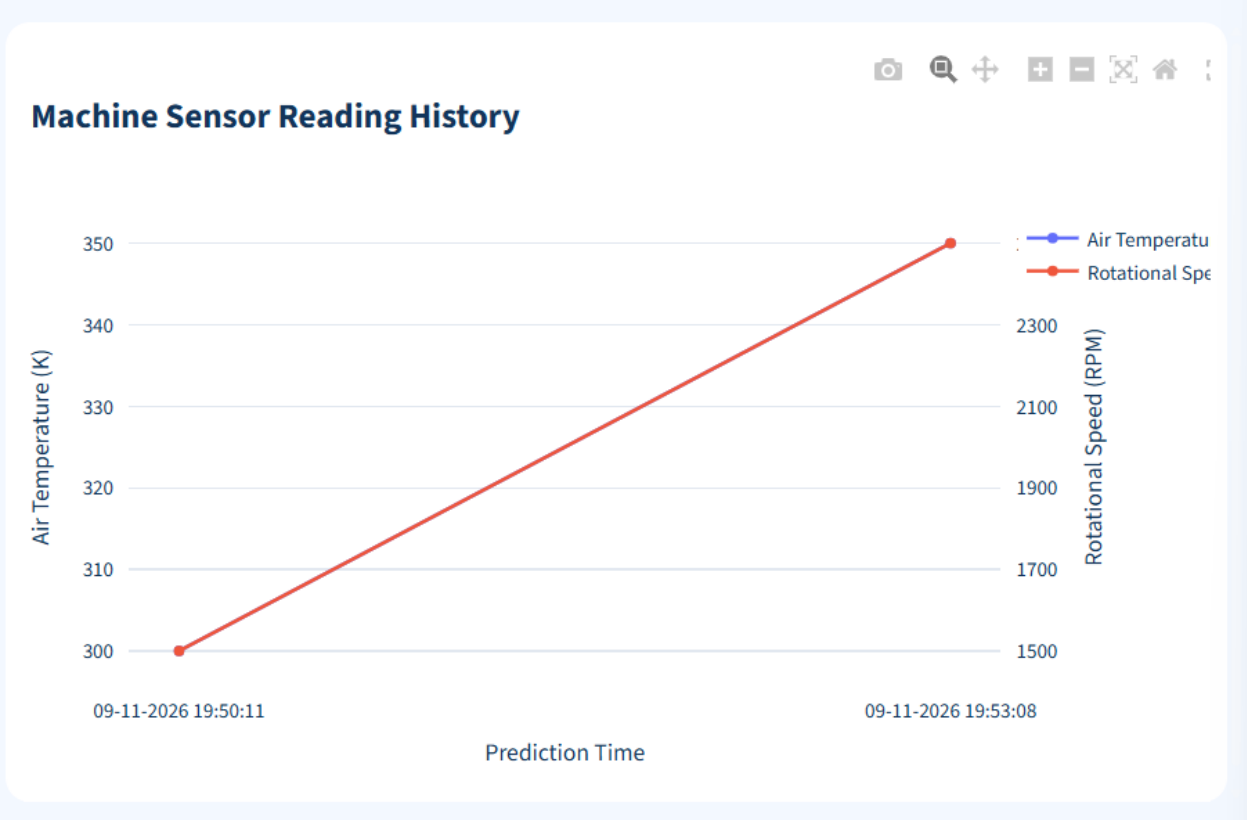


11.2 Sensor Reading Visualisation

An interactive graph was developed using Plotly to visualise machine sensor readings over prediction history. It displays changes in air temperature and rotational speed.

This visualisation helps users understand variations in machine operating conditions.

Figure 10: Machine Sensor Reading History



11.3 Overall Results

The Random Forest model achieved 98.15% accuracy on the testing dataset and was selected for integration into the Streamlit application.

The application successfully performed both failure and no-failure predictions during testing. Prediction history, sensor visualisation and CSV download features were also tested successfully.

The project demonstrates the practical use of machine learning for identifying potential industrial machine failures.

12. ADVANTAGES AND LIMITATIONS

12.1 Advantages

- Helps identify potential machine failures using sensor data.

- Reduces dependence on traditional fixed maintenance schedules.
- Compares multiple machine learning algorithms.
- Provides an easy-to-use web application.
- Supports prediction history and CSV data download.
- Helps improve maintenance planning and operational awareness.

12.2 Limitations

- The model is trained using a synthetic dataset.
- Prediction performance may differ with real industrial machine data.
- The system provides binary failure predictions rather than identifying the exact failure type.
- The model does not predict the exact time remaining before a machine failure.
- Real-time sensor integration is not currently implemented.

13. CONCLUSION AND FUTURE ENHANCEMENTS

This project successfully demonstrates an AI-based predictive maintenance system for industrial machines using machine learning techniques.

Three algorithms, namely Logistic Regression, Decision Tree and Random Forest, were implemented and compared. Random Forest achieved the best overall accuracy of 98.15% and was selected for the final application.

A user-friendly Streamlit application was developed to accept machine sensor readings and display failure predictions. Additional features such as prediction history, sensor visualisation and CSV download were also implemented.

In the future, the system can be improved by integrating real-time industrial sensors, using real-world machine datasets, predicting specific failure types and estimating the remaining useful life of machines.

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